

Jacob Edwards, PhD

Data Scientist

jacobedwards.jae@gmail.com
(865) 771-9750
portfolio: <https://amnion.github.io/>

Data scientist with 13 years of experience designing experiments and testing models. Specialized in handling time-series signals and other unstructured data. I am action-oriented, excel in ambiguous, fast-paced environments, and approach analytics with disciplined reasoning cultivated in scientific rigor. My goal is to help teams make informed decisions by discerning signal from noise.

SKILLS

Programming & Tools

Python, MySQL, R, MATLAB, MS Office

Data Analysis

Statistical modeling, regression and classification, time series analysis, natural language processing, machine learning, feature engineering, dimensionality reduction, model evaluation for explanation and prediction, data visualization

Experimentation

Hypothesis testing, experimental design, effect size estimation, uncertainty quantification, bootstrapping, simulation

Communication

Data storytelling, technical writing, stakeholder communication, cross-functional collaboration, teaching and mentoring

RELEVANT EXPERIENCE

Research Scientist

COLUMBIA UNIVERSITY

New York, NY

2019 - 2025

Led end-to-end execution of research projects involving large-scale time-series and sequential signal data, from question and hypothesis formulation to experimental design and data collection, through to modeling, validation and delivery to stakeholders.

Time-Series Modeling & Workflow Automation

- Generated and analyzed 3.2 TB of stimulus-response time-series neural data. Built scalable workflows for signal cleaning, feature extraction, aggregation and visualization.
- Replaced a manual data quality screening workflow with a supervised ML-based [anomaly detection classifier](#), reducing manual review time by ~20% and improving consistency across projects.
- Implemented statistical and predictive models, including generalized linear models to estimate spectrotemporal receptive fields, enabling the quantitative comparison of neural responses across conditions.
- Built [interactive dashboards](#) in Python (Dash) and MATLAB to support exploratory analysis and enable non-technical collaborators to interpret multi-dimensional analytical results.

Sequence Modeling & Structured Data Engineering

- Generated and analyzed large-scale acoustic datasets comprising more than 24,000 vocal events across 138 subjects. Optimized event segmentation pipeline to derive structured tabular representations from raw signals.
- Engineered features from audio recordings using domain metrics and decomposition techniques (PCA, wavelets). Applied dimensionality reduction and clustering techniques to identify latent structure in vocal events.
- Created a [graphical user interface](#) to unify database management and data entry pipelines. The GUI trivialized manual annotation (reduced work hours per project ~80%) and enabled non-technical interns to collaborate on project progress.
- Developed and evaluated NLP model for sequence prediction using non-lexical signal features. Designed a custom cost metric based on sequence distance algorithms. Created custom effect size metrics and bootstrapped confidence intervals.

Leadership & Cross-Functional Impact

- Secured \$140,256 in competitive [federal research funding](#) by translating technical proposals into clearly articulated project plans and measurable outcomes.
- Developed and taught an original, [applied data science curriculum](#) (20 hours of lecture, 7 homework assignments) covering statistical modeling, data wrangling, visualization and machine learning. Emphasized the translation of real-world data analysis for policy decision-making and the delivery of actionable insights to stakeholders.
- Authored 6 peer-reviewed publications in high-impact, general and specialist journals including *Nature Scientific Reports* and the *Journal of Neuroscience*.

SELECTED PUBLICATIONS

2026	Syntax models of species' rules accurately predict vocal sequences	[in press]
2025	Temporal organization of behavior is driven by species, not learning	paper
2023	Machine-learning analysis of song links vocal acoustics to ancestry	paper

EDUCATION

PhD	2025	Columbia University
MSc	2018	Colorado State University
BSc	2013	Uni. of Tennessee, Knoxville